

Question_8:

Code:

```
1  /*
2  Question No: 8
3  For every day, an age is given as the Ghost of the Day.
4  After each day, find:
5  1. The age of the ghost who has the most titles.
6  2. The number of titles won by that ghost so far.
7  If two ghosts have the same number of titles,
8  the older ghost wins.
9  */
10
11 #include <iostream>
12 #include <map>
13 using namespace std;
14
15 int main(){
16     int n,m;
17     cin>>n>>m;
18
19     map<int,int> cnt;
20     for(int i=0;i<n;i++){
21         int x;
22         cin>>x;
23
24         cnt[x]++;
25         pair<int,int> best={-1,-1};
26
27         for(auto p:cnt)
28             if(p.second>best.first ||
29                (p.second==best.first && p.first>best.second))
30                 best={p.second,p.first};
31         cout<<best.second<<" "<<best.first<<"\n";
32     }
33 }
```

Output:

Test Case_1:

```
7 5
1 3 1 3 2 2 2
1 1
3 1
1 2
3 2
3 2
3 2
2 3
PS C:\Users\Prabin Yadav\
```

Test Case_2:

```
10 4
3 4 5 1 3 1 3 2 2 2
3 1
4 1
5 1
5 1
3 2
3 2
3 3
3 3
3 3
3 3
PS C:\Users\Prabin Yadav\
```